

The rate-utility theorem

August 2024

Theorem 1. Fix a utility function $u: \mathcal{X} \times \mathcal{Z} \rightarrow \mathbb{R}$. Fix an input random variable X , i.e., with domain $\mathcal{X}$. For a noisy channel with capacity c used at rate r , the supremal expected average utility over coding schemes of input sequences $x^{1:n}$ to output sequences $z^{1:n}$ through that channel is

$$\sup_{\text{joint distributions on } X \text{ and } Z \text{ marginalizing to the given distribution on } X \text{ and with } I(X;Z) < rc} \mathbb{E}_{p(x,z)} [u(X, Z)].$$

Remark. One can also replace the expected average in coding with a whp (or exponentially good) lower bound on the average.

Proof. To go from coding to random variables, use the coding as random variables together with an index trick.

To go from random variables to coding, sample $2^{n(I(X;Z)+\epsilon)}$ sequences $z^{1:n}$ and for each $x^{1:n}$, just send whichever one has the highest utility. To see that this gives utility at least as good as the expected utility when $z^{1:n}$ is sampled, we will restrict to only the case where the sequence $(x^{1:n}, z^{1:n})$ is ‘typical’, ie has probability in the range $2^{-n(H(X,Z)\pm\delta)}$ and has prob of $z^{1:n}$ given $x^{1:n}$ in the range $2^{-n(H(Z|X)\pm\delta)}$ (all this happens except with exponentially small probability). Call an $x^{1:n}$ ‘typical’ when it has exponentially high probability of producing a typical $(x^{1:n}, z^{1:n})$ ($x^{1:n}$ being typical in this sense again happens except with exponentially small probability).

Now fix any typical $x^{1:n}$. We will see that with exponentially high probability, something in the size $2^{n(I(X;Z)+\epsilon)}$ selection will beat whatever gets sampled conditional on $x^{1:n}$. I guess the key observation of this argument is this: for each typical $(x^{1:n}, z^{1:n})$, the probability that this $z^{1:n}$ is included in the selection of $2^{n(I(X;Z)+\epsilon)}$ sequences is exponentially higher than the probability it is picked conditional on this $x^{1:n}$. Given this, we just look at an exponentially small probability highest-utility tail of the typical sequences (everything up to the first sequence which makes the total prob larger than some exponentially small quantity), and note that the prob that the conditionally picked $z^{1:n}$ is strictly in this tail¹ is exponentially small (by def), whereas the prob that one thing in the selection is in this tail can be made exponentially close to 1 (here one of course needs to calculate). So whenever all this stuff happens together, the selection does better than the conditionally drawn $z^{1:n}$. So the selection beating the test channel has probability exponentially close to 1. This also implies that the expected utility of the coding scheme is larger than the expected utility of the test channel modulo an exponentially small term.

Ok yes, this had the rate be a tiny bit larger than what we’d want. To get rate $I(X, Z)$ instead of $I(X, Z) + \epsilon$, I guess we can just settle with encoding a length $(1 - \epsilon)n$ sequence and noting that this error can again be made arbitrarily small. (If one cares, one can really probably sacrifice much less than a const fraction here, and in fact I think get this utility error term to be exponentially small like the other error term.) $\square$

¹ie in the tail but not the very last thing put in the tail